CCCA Cell-State Analysis · Griffiths et al. 2021 — Breast Cancer

GO / NO-GO GATES

Gate 1 — Integration & Continuity — GO

Harmony on `sample` (theta sweep 2/10/20; per-neighbourhood iLISI 1.7 -> 19.2 at theta=10, the peak). The regenerated embedding is one continuous, thoroughly sample- and patient-mixed body; cell cycle is flat (no proliferation lobe). Only a 328-cell iCell8 technology streak + tiny specks detach (island removal 110,195 -> 109,353). Mixed AND continuous.

Gate 2 — Module Geography — GO (non-branching; FLAG)

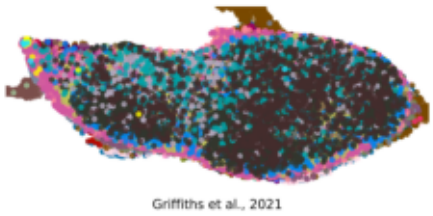
Organised Gast<->pSom cross-repression along one fold-free axis (pseudotime rho: Gast +0.40, pSom -0.36) with a terminal Crest gradient (rho +0.28) — the non-branching pattern of the task spec. FLAG: Crest enriches the Gast-leaning terminus, not pSom (an inversion; passes anyway).

Gate 3 — Cross-repression (branching only) — NA

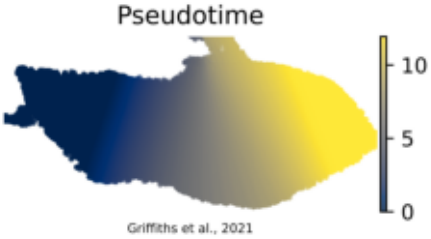
No bifurcation: a PPT tree yields 15 tips (13 branch nodes) all with identical module identity, and Gast/pSom co-express below the independence expectation (7.3% vs 9%) — no convergence hub. Gate 3 is branching-only, so NA.

STEP FIGURES

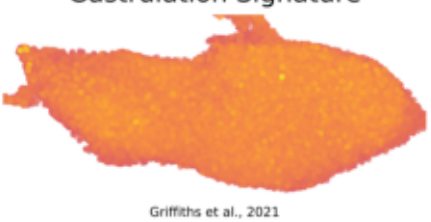
Gate 1 — sample integration



Trajectory — pseudotime (root left)



Gastrulation signature



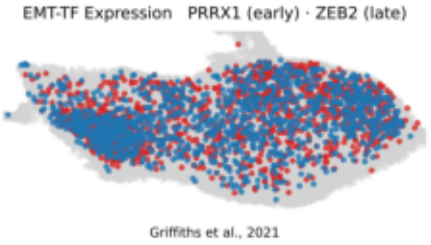
Somitogenesis signature



Neural Crest signature



EMT-TF expression



Dataset Breast cancer (ER+ neoadjuvant letrozole ± ribociclib) · 10x + iCell8 · 109,353 malignant cells, 67 samples.

Verdict

The embryonic programme reproduces as a COLLAPSED bifurcation: a single continuous gastrulation/somitogenesis <-> neural-crest axis (geometry class Linear-pSom/Crest; module means Gast -0.019, pSom -0.053, Crest +0.023). Gast and pSom cross-repress along the axis but never resolve into two competing arms, so there is no hub and no Gate-3 test. Crest is terminal but enriches the Gast side (flag). EMT-TFs are near-silent in this treated luminal cohort; the most detectable pseudotime-graded pair is PRRX1 (root) <-> ZEB2 (terminus).

Key parameter choices

Min-cell threshold	>= 100 cells/sample (gap-finder degenerate on smooth dist) -> 67/95 samples
HVG / n_pcs	4,072 over-dispersed genes; n_pcs = 20 (scree elbow ~PC7, ~32% cum. var)
Integration	Harmony on `sample`, theta = 10 (iLISI peak 19.2; theta=20 over-integrates)
Island removal	6x-median radius graph, largest component (842 cells / 57 specks removed)
Geometry	Non-branching fold-free curve (fold ratio 2.95); root = Gast/pSom pole
Root / EMT-TFs	pSom-high developmental origin; PRRX1 (early) <-> ZEB2 (late)
Hub identity / r	n/a — no convergence hub (Gate 3 NA)